EXPERIMENT 7: IMAGE CAPTIONING USING CNN-RNN ARCHITECTURE

Aim:

Generate a meaningful caption for an image using a pre-trained Vision Transformer (ViT) image encoder and GPT-2 text decoder.

Procedure

1. Import PyTorch, Transformers, and PIL.

2. Load the pre-trained ViT-GPT2 image-captioning model, processor, and tokenizer.

3. Select CPU or GPU and load the model.

4. Define a function to preprocess an image and generate a caption.

5. Generate text using beam search decoding.

6. Supply an image named download.jpeg and print the caption.

Cell 1 – Import required libraries

```
import torch
from transformers import VisionEncoderDecoderModel, ViTImageProcessor, AutoTokenizer
from PIL import Image
```

Result: The required libraries for image processing and caption generation are imported.

Cell 2 – Load the pre-trained image-captioning model

```
model_name = "nlpconnect/vit-gpt2-image-captioning"

model = VisionEncoderDecoderModel.from_pretrained(model_name)
processor = ViTImageProcessor.from_pretrained(model_name)
tokenizer = AutoTokenizer.from_pretrained(model_name)

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)
model.eval()
```

Result: The pre-trained model, image processor, and tokenizer are loaded.

Cell 3 – Define the caption-generation function

```
def generate_caption(image_path, max_length=30, num_beams=4):
    image = Image.open(image_path).convert("RGB")
    pixel_values = processor(
        images=image, return_tensors="pt"
    ).pixel_values.to(device)

    output_ids = model.generate(
        pixel_values,
        max_length=max_length,
        num_beams=num_beams,
        early_stopping=True
    )
    return tokenizer.decode(output_ids[0], skip_special_tokens=True)
```

Result: A function is created to preprocess an image and generate a caption.

Cell 4 – Load an image and generate its caption

```
if __name__ == "__main__":
    img_path = "download.jpeg"  # Put an image with this name in the working folder
    print("Caption:", generate_caption(img_path))
```

Result: The model prints a caption describing the supplied image.

Output

After the model and image load successfully, output appears in this format. The caption depends on the image used:

```
Caption: a person riding a bicycle on a road
```

RESULT

The pre-trained ViT-GPT2 image-captioning model was used to generate a natural-language caption for an input image. The caption describes visual content detected by the model.